

Lab 3: Component Modelling & Architectural Pattern Selection

Written Architectural Justification — Self-Service Coffee Kiosk System

Architecture Selection: We chose **Layered Architecture** for the **Self-Service Coffee Kiosk System**.

Architectural Choice

The system is organized into three logical layers: **Presentation**, **Business**, and **Data**. The Presentation Layer handles the touchscreen user interface; the Business Layer contains the Order Manager, Payment Service, and Receipt Printer interactions; and the Data Layer stores menu and pricing information. This separation keeps user interaction, order processing, and data management as distinct responsibilities.

Two Scenario-Specific Reasons

No.	Reason	Justification
1	Clear separation of responsibilities	The kiosk has distinct concerns: touchscreen ordering, order/payment processing, receipt printing, and menu/pricing storage. Layering separates these concerns, making the component interactions easier to understand and maintain.
2	Suitable for a focused kiosk application	The kiosk performs a relatively well-defined ordering workflow. A layered design provides a straightforward structure without the operational complexity of independently deploying and coordinating many microservices.

Security Advantage

Layer separation allows access to payment and stored menu/pricing data to be controlled through defined component interfaces rather than allowing the touchscreen layer to access data directly. The Payment Service can therefore receive controlled payment-processing requests from the Order Manager, while database access remains within the Data Layer.

Performance Benefit

For this focused kiosk workflow, the layered design can keep the main ordering logic within the application rather than requiring communication between many independently deployed services. This avoids unnecessary distributed-service network overhead for routine operations such as selecting a drink, checking pricing data, and completing an order.

Conclusion: Layered Architecture provides a clear structure for the kiosk by separating touchscreen interaction, business processing, and menu/pricing data while keeping component communication controlled through defined interfaces.